# Chapter 19 Notes

## Section 2

### Notes

For the model in this section to work, we need $\omega_2 \in M$. However, it is not clear to my why this is.

If we assume that $\omega_2 \in M$, here is why $\langle \mathcal{G}, \mathcal{R} \rangle$ is an $M$-set. It is easy to see that each particular forcing condition is in $M$ - each particular one can be explicitly defined, as it is assumed that $p(\alpha) = \langle \emptyset, \emptyset \rangle$ for all but a finite number of $\alpha \in \omega_2$. The $\mathcal{G}$ set of all forcing conditions is a subset of $\mathcal{P}(\mathcal{P}(\mathcal{P}(\omega_2 \times \mathcal{P}(\omega \times \omega) \cap M) \cap M) \cap M) \cap M$, which is in $M$. $\mathcal{G}$ is also easily seen to be definable over $M$, and consequently, $\mathcal{G} \in M$. Then $\mathcal{R}$ is also easily seen to be in $M$, as it is definable over $M$, and is a subset of $\mathcal{G} \times \mathcal{G}$, which is in $M$ (as we have shown $\mathcal{G}$ to be in $M$).

### Exercise 2.1

This is basically the same proof as Theorem 4.1 of Chapter 18.

*Proof.* We prove that, for each $\alpha \in \omega_2$, $[\![\sim\text{Const}(f_\alpha)]\!]$ is valid in the model $M$. Suppose, to the contrary, that $p \nvDash [\![\sim\text{Const}(f_\alpha)]\!]$ for some $p$. Then for some $p'$ with $p \mathcal{R} p'$, $p' \vDash [\![\text{Const}(f_\alpha)]\!]$. By Theorem 3.2 of Chapter 18, it follows that $p \vDash \Diamond [\![f_\alpha \approx \hat{c}]\!]$ for some contrsuctible set $c \in M$. This means that, for some $p''$ with $p' \mathcal{R} p''$, $p'' \vDash [\![f_\alpha \approx \hat{c}]\!]$. We derive a contradiction by showing $p'' \vDash [\![\sim(f_\alpha \approx \hat{c})]\!]$. $p''$ is some forcing condition, and suppose that $p''(\alpha) = \langle P, N \rangle$. There are then two cases.

Case 1) $c \subseteq P$. Because both $P$ and $N$ are finite, we can pick some integer $n$ such that $n \notin P \cup N$ (thus also $n \notin c$). Let $P' = P \cup \{n\}$. Then there is some forcing condition $q$ such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = p''(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P', N \rangle$. Obviously $p'' \mathcal{R} q$. Because $c \in P'$, it follows that $q \vDash [\![\hat{n} \in f_\alpha]\!]$. But because $n \notin c$, $q \vDash [\![\sim(\hat{n} \in \hat{c})]\!]$. These two things jointly imply $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$, as $p'' \vDash [\![f_\alpha \approx \hat{c}$ and $p'' \mathcal{R} q$. Thus this is our contradiction in this case.

Case 2) $c \nsubseteq P$. Pick some $n$ in $c$ but not in $P$. Let $N' = N \cup \{n\}$. Then there is some forcing condition $q$ such that, for all $\beta \in \omega_2$, if $\beta \neq \alpha$, then $q(\beta) = p''(\beta)$, and if $\beta = \alpha$, then $q(\beta) = \langle P, N' \rangle$. Obviously $p'' \mathcal{R} q$. Because $n \in c$, it follows that $q \vDash [\![\hat{n} \in \hat{c}]\!]$. But because $n \in N'$, it follows that $q \vDash [\![\sim(\hat{n} \in f_\alpha)]\!]$. These two things imply that $q \nvDash [\![f_\alpha \approx \hat{c}]\!]$. But also $q \vDash [\![f_\alpha \approx \hat{c}]\!]$ (for the same reason as in the first case), which is our contradiction in this case.

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